
Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

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Abstract

Quantizing a large language model reduces serving cost and costs some accuracy. Determining how much normally requires re-running a full benchmark suite, which is the expense quantization was meant to avoid. We ask whether the accuracy delta can be predicted instead. Using 817 published evaluations spanning 38 base checkpoints, 8 model families and 6 quantization schemes, we report three results. First, a *negative* one: per-model prediction carries almost no signal. Under leave-one-family-out validation the best predictor, a per-scheme mean, beats a global mean by 0.03pp of MAE, and both ridge regression and gradient boosting are *worse* than guessing the average, because evaluation noise accounts for at least a third of the variance on 8 of 11 benchmarks. Second, a positive one: conditioning only on the quantization scheme yields split-conformal intervals with 90.1% empirical coverage (118/131, 95% CI [83.6, 94.6]) on checkpoints never used in construction. Third, and most consequential for practice: the published corpus is outcome-truncated. Every configuration in it is one that someone chose to release. We rented an A100 and measured deliberately-faulted configurations, observing losses to -39.5pp against a published worst case of -8.86pp —measured, we stress, only on models of 1.5B and below, and on configurations engineered to fail rather than typical ones. We argue the resulting artifact should be described as a calibrated historical baseline rather than a predictor, and we state precisely why its coverage guarantee does not transfer to a practitioner’s own untuned recipe. Code, data and the provenance of every reported figure are provided as supplementary material.

1 Introduction

Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8 wants one number: how much accuracy will this cost? In practice the only way to know is to run the evaluation.

This paper asks whether that number can be forecast from the public record. Model publishers release quantized checkpoints with benchmark tables comparing the compressed model against its dense parent. Aggregated, these are hundreds of paired measurements of exactly the quantity of interest (817 in our harvest).

Our contributions:

1. **A negative result (§5).** Per-model prediction of the accuracy delta does not work, and we show why: the target is dominated by evaluation noise. We report this as the primary finding rather than burying it.

2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured coverage on held-out families and an explicit rule that withholds an interval where coverage is poor or the evidence is too thin to judge.
3. **A measurement of the corpus’s own selection bias** (§7), via GPU experiments on deliberately-faulted configurations—data the published record structurally cannot contain.
4. **An audit** (§9) recording errors we found in our own work, including ones that changed published numbers and claims.

2 Related Work

Predicting evaluations. BenchPress Zeng and Papailiopoulos [2026] establishes the mechanism closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is a model’s absolute score; ours is the paired delta between a model and its compressed self, and quantized checkpoints do not appear in their paper.

Conformal prediction on compressed models. Tong et al. Tong et al. [2026] apply conformal prediction directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The estimand differs: their conformal sets are over label space, constructed from a compressed model’s own output probabilities, so the method measures uncertainty *after* running the model. Ours is constructed over historical accuracy deltas to give an interval *before* running anything. Their Figure 2 plots the same quantity we predict, $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$, but measures it. Their finding that compression decouples accuracy from uncertainty bounds what an accuracy-only estimate like ours can claim (§8).

Quantization methods. The checkpoints we harvest are produced with GPTQ Frantar et al. [2022] and SmoothQuant Xiao et al. [2022] via llm-compressor vLLM Project [2024].

The source of our data. Kurtić et al. Kurtić et al. [2024] report over 500,000 evaluations of quantized Llama-3.1 models, finding recovery above 99% of the average OpenLLM v1 score for every scheme tested. As far as we can tell, that work does not address whether the published set of quantized models is a biased sample. Characterising that gap is the contribution of §7.

Selection bias. Survivorship bias has been identified and measured in a widely used retrieval benchmark Gupta and MacAvaney [2022]. Conformal prediction’s guarantee is known to fail under selection: conditional on a selection event, calibration data are no longer exchangeable with a test point Barber et al. [2023], Jin and Ren [2024]. We apply this directly to our own artifact in §8.

3 Data

We harvest published model cards into rows of (model, scheme, benchmark, $\text{acc}_{\text{before}}$, $\text{acc}_{\text{after}}$), yielding 817 rows after filtering, across 38 base checkpoints, 8 families and 6 schemes.

An integrity gate. Cards typically print a *Recovery %* column alongside the before/after pair. This is redundant information, and we exploit it: for every three-column row we check $\text{recovery} \approx 100 \times \text{after}/\text{before}$ to a tolerance derived from the card’s own printed precision, and reject rows that disagree. This caught spurious values that our parser produced under column-order variation between card templates, and internally inconsistent cards. Rejections are recorded rather than silently dropped.

A structural limitation, stated at the outset. All rows come from a single publisher using a single toolchain. Holding out a model family tests transfer across *models*; nothing in this design tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

79 4 Method

80 **Split conformal.** Given calibration residuals $\{s_i\}_{i=1}^n$ and level α , the interval half-width is the
 81 $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which gives exact finite-sample coverage under exchangeability.
 82 Where n is too small for that index to exist, we return ∞ rather than a misleadingly finite band.

83 **Mondrian (group-conditional) calibration.** A single marginal interval advertised at 90% was
 84 in fact a 68% interval for NVFP4 users—the users who most need it. We therefore calibrate per
 85 scheme.

86 **A three-state verdict.** For each (scheme, size-band) cell we measure coverage directly, and the
 87 tool returns one of three verdicts. A cell is *trusted* when measured coverage is adequate and ad-
 88 equately supported. It is *refused*—INSUFFICIENT CALIBRATION, no interval—when coverage is
 89 measurably poor on a sufficient number of scored rows. It is *insufficient evidence* when the cell rests
 90 on fewer than 3 distinct checkpoints, or when a checkpoint-level cluster bootstrap 90% interval on
 91 its coverage straddles the 85% line. In that case the evidence cannot support either of the other two
 92 verdicts. The tool still prints the all-sizes interval for such a cell, but demotes the scheme to the
 93 lowest tier and states that the cell cannot be judged; only a refused cell has its interval withheld.

94 Coverage is scored *one-sided*: how often the true delta stayed at or above the interval’s lower bound.
 95 The risk this tool bounds is accuracy loss, and a model that beats its envelope has exposed no one
 96 to anything, so counting that as a failure would flag cells that are merely outperforming. Comparing
 97 a one-sided measurement against the printed 90% is the *lenient* reading, not a flattering one: the
 98 interval is symmetric, so the permitted 10% is split across two tails, and pooled one-sided coverage
 99 across all scored rows is 95.1%.

100 Two cells were flagged under the earlier two-state rule, and neither is a confident refusal under the
 101 three-state one. `w4a16|<2B` had one-sided coverage of 70.1% over 77 rows (two-sided 68.8%), but
 102 those rows come from only 2 checkpoints. `w8a16|>10B` had 81.1% over 217 rows from 5 check-
 103 points (two-sided 73.7%), and its bootstrap interval, $[47, 100]\%$, straddles the threshold. Both are
 104 now *insufficient evidence*. We had scoped the change to those two cells; applying the rule to every
 105 cell moved 8 of 17 into that state, including several whose point-estimate coverage looked comfort-
 106 able. The one-sided scoring and the three-state rule were proposed by an external commenter who
 107 identifies as an AI account collaborating with the OpenLanguageModel maintainers; we recomputed
 108 each of their figures from the raw rows before adopting either change, and one of their descriptions
 109 did not survive that check.

110 **A support floor.** Coverage measured on many benchmarks from one checkpoint is not indepen-
 111 dent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct
 112 checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

113 5 The negative result

114 Under leave-one-family-out validation, row-weighted MAE:

	predictor	MAE (pp)	beats global mean?
	per-scheme mean (shipped)	0.7227	yes
	global mean (baseline)	0.7545	—
115	per-(scheme $\times$ benchmark)	0.7607	no
	ridge regression	0.7639	no
	per-benchmark mean	0.7647	no
	gradient boosting	0.7878	no

116 Ridge and gradient boosting are both *worse* than predicting the global average. Adding model size,
 117 bit-width, method, benchmark identity and headroom each degraded out-of-family performance.

118 **Why.** Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta
 119 should be approximately zero—gives a noise floor that accounts for anywhere from 15% of the

120 observed variance to all of it across 11 benchmarks. On GSM8K the deltas of supposedly lossless
 121 schemes have a standard deviation of 2.1pp, while the quantity being predicted has mean -0.39pp .
 122 The target is a difference of two noisy quantities that largely cancel: the numerically smallest and
 123 hardest-to-predict summary of quantization damage available.

124 **Implication.** The entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band,
 125 with no model and no conformal machinery, is a serious competitor: it looked better covered in-
 126 sample, though under the strict protocol that advantage disappears (§9). What conformal contributes
 127 is a finite-sample guarantee and a principled rule for small- n schemes, not accuracy. We therefore
 128 describe the result as a *calibrated historical baseline*, not a predictor.

129 6 Results

	scheme	90% interval (pp)	worst observed	% losing $>3\text{pp}$	evals
	fp8	$[-1.11, +0.48]$	-1.90pp	0.0%	80
	w8a8_int	$[-1.71, +1.07]$	-3.32pp	1.0%	194
130	fp8_dynamic	$[-1.44, +1.34]$	-8.72pp	1.0%	200
	w8a16	$[-0.98, +0.81]$	-3.04pp	1.2%	86
	w4a16	$[-2.87, +1.40]$	-8.86pp	5.7%	193
	nvfp4	$[-4.10, +1.87]$	-8.12pp	15.6%	64

131 **Prospective validation.** After freezing the method we harvested checkpoints not used during de-
 132 velopment. Under a strict protocol that excludes any family present in training and any card whose
 133 format informed a parser fix, coverage is $118/131 = 90.1\%$, 95% Clopper–Pearson CI $[83.6, 94.6]$,
 134 against a nominal 90%. A from-scratch, standard-library-only reimplementation sharing no code
 135 with the pipeline reproduces this figure with zero disagreements.

136 **What the envelope cannot do.** 0 of 6 intervals exclude zero: the tool can never state that a scheme
 137 *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when
 138 real damage occurs, the interval usually misses it. This is the single most important caveat in the
 139 paper.

140 7 The measured left tail

141 The corpus contains only configurations someone chose to publish. To measure what it structurally
 142 cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

143 **Result.** Worst measured loss -39.5pp , against -8.86pp as the worst loss anywhere in the pub-
 144 lished corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-
 145 configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor
 146 instead of per-group.

147 **Scope, stated plainly.** This tail was measured only on models of 1.5B and below. It establishes
 148 that catastrophic damage is reachable; it is not a measured bound for larger models.

149 **A confound we found and did not resolve.** Re-running MMLU under the published protocol
 150 reproduces published base accuracy (47.20 against 47.42; 59.57 against 60.98), closing the harness
 151 question. It exposed a larger one: our own *correctly-configured* control is $2.6\times$ more damaging
 152 than published W4A16 on closely comparable checkpoints (mean gap -3.78pp , $z = 2.9$ and 5.3).
 153 We attribute this to approximately $171\times$ less calibration data than production defaults; we found
 154 no bug. Consequently, excess-over-control measures *bad recipe versus mediocre recipe*, not versus
 155 production. It is a conservative anchor, not a bound, and we decline to publish a corrected point
 156 estimate: the correction moves by 5.70pp depending on the anchor set, and we have only 6 rows
 157 from two models.

8 Limitations

The guarantee is conditional, and absent where it matters most. Split conformal gives exact coverage under exchangeability between calibration and test. Our calibration set is the set of configurations a publisher chose to release—a selection event. Conditional on selection, calibration points are not exchangeable with an arbitrary test point and the guarantee does not transfer Barber et al. [2023], Jin and Ren [2024]. It holds for a new checkpoint drawn from the same publication process. It carries **no guarantee** for a recipe the reader tuned themselves and that no publisher would have released—which is the case a practitioner most wants covered.

Accuracy is not the whole of deployment risk. Tong et al. [2026] show compression frequently decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets inflate. A scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We measure none of that.

One publisher, one toolchain, and why. See §3. Our own control arm (§7) is direct evidence that practitioner and calibration volume can dominate scheme choice. We measured how confining this is: surveying 10 publishers and 260 quantized model cards, 10 of 222 cards from the other 9 publishers carry paired before/after evaluations, 9 of them Intel’s, so others do publish them. What the other publishers almost never provide is a *Recovery* column, which is what lets each row be checked against 100 $\times$ after/before and rejected on disagreement: of their rows, 6 pass that gate, all from 1 card, against 147 from the control publisher. The corpus is single-publisher not because others publish nothing, but because the control publisher’s cards are, by a wide margin, the ones published in a self-verifying form.

Rows are not independent. Up to sixteen benchmark rows come from a single quantization run. We mitigate with group-conditional calibration and the support floor, and note that a cluster bootstrap over model configurations does not materially widen the headline interval, but the unit of analysis remains a row.

9 Audit

We record errors found in our own work, including those that changed published claims. This section exists because the alternative—reporting only what survived—would misrepresent how the result was obtained.

Errors that changed a published number.

- *A retracted headline.* An early claim that held-out models came from families never used in construction was wrong; they were new checkpoints of families already present. The strict figure is 118/131.
- *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was matched as Math-Lvl-5, contaminating 7 test rows with scores from a different benchmark.
- *A centring mismatch.* Calibration measured residuals about the stratum mean while intervals centred on the scheme mean, mis-sizing every calibrated width. Found only by independent reimplementations.
- *A column-order fault.* One card template orders columns [recovery, base, quant], which our parser turned into a spurious -68pp delta. The integrity gate rejected 10 of 11 rows on that card, passing only the one where base = quant made the check degenerate.
- *A retracted single-publisher claim.* We had asserted that one publisher was close to the only source of paired evaluations. A census of 10 publishers found 10 paired-evaluation cards outside the control publisher, 9 of them Intel’s; what differs is whose rows can be verified (§8).
- *A bootstrap on the wrong statistic.* The checkpoint-level bootstrap in our coverage audit re-sampled the two-sided indicator while the verdict was judged on the one-sided statistic. An

external commenter asked for it to wrap the one-sided figure, which exposed the mismatch (§4).

A regression we nearly shipped. An audit indicated that a raw asymmetric empirical band covered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch. On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-sided empirical index requires more calibration points than are often available. An infinite interval covers by construction. On rows where both are finite, the original wins on coverage *and* width. The change was reversed.

An overclaimed guarantee. The artifact was described as carrying a coverage guarantee without qualification. Checking the method against the conformal literature rather than against our own earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected in §8.

Provenance discipline. Two parser fixes were informed by inspecting rows that were already in the test set. Rather than silently retain them, we record which fixes were test-informed and exclude the affected family from the strict headline.

Mechanical verification. The figures this paper reports as results are macros generated from a claims registry that recomputes each value from source artifacts, and the accompanying code fails a test if an outward-facing document contains an untraced numeric literal. The exceptions are historical figures in the error list above (row counts and the -68pp delta) and the 90% marginal-interval figure in §4, which are quoted from the provenance record rather than recomputed.

10 Conclusion

Per-model prediction of quantization accuracy damage does not work on published data, and the reason is largely measurement noise rather than model capacity. What does work is narrower: a per-scheme calibrated envelope, validated at 90.1% coverage on unseen checkpoints, that declines to answer where its own measured coverage is poor or too thinly supported to judge. The more useful contribution may be the limitation itself—the published record of quantized models is outcome-truncated, and we measured, on deliberately faulted recipes for models up to 1.5B, losses more than four times worse than anything it contains. A risk estimate built on that record is a floor on risk, not a ceiling, and should say so.

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